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Generation and characterization of bio-oil obtained from the slow pyrolysis of cooked food waste at various temperatures

Sourodipto Modak [a], Priyanka Katiyar [a,*], Sanjeev Yadav [a], Siddharth Jain [a], Bappaditya Gole [b], Dhrubajyoti Talukdar [b]

[a] Department of Chemical Engineering, Shiv Nadar Institution of Eminence (Deemed to be University), Gr. Noida 201314, Uttar Pradesh, India
[b] Department of Chemistry, Shiv Nadar Institution of Eminence (Deemed to be University), Gr. Noida 201314, Uttar Pradesh, India

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ABSTRACT

Bio-oil was generated from slow pyrolysis of cooked food waste (CFW) at various temperatures (300–500 °C). Then NMR analysis was used as a qualitative means to characterize the bio-oil for its nature (aliphatic or aromatic), and then the compounds were confirmed and quantified using the GC–MS. This analysis indicated that the pyrolysis at low temperature (300 °C) mainly generated carbonyl compounds (Aldehydes, Ketones, Esters, and Oxo groups), Levoglucosans, and Furans (17%, 24%, and 38%, respectively) considered as typical pyrolysis chemicals. Similarly, the pyrolysis at medium temperature (400 °C) generated other compounds that were present in significant quantity, including sugars, aliphatic compounds, nitrogen compounds, acids, phenolic compounds, and alcohols. However, their fraction decreased with an increase in pyrolysis temperature to 500 °C and the fraction of aromatics increased significantly (>60%). This aromatics fraction was much more than that in a bio-oil from typical biomass which can be attributed to distinctively different chemical characteristics of CFW due to presence of additional compounds such as starch, proteins, waxes and oils in CFW. Moreover, the composition of aromatic fraction was better because a very high percentage of aromatic ethers (>58%) e.g. Benzene, 1,3-bis (3-phenoxyphenoxy), was found at 500 °C which can be converted into aliphatic alkanes, aliphatic alcohols, aromatic derivatives and platform chemicals by means of catalyst addition.

1. Introduction

The generation of cooked food waste (CFW) is a pressing matter in modern times. India generates around 68,760,163 tonnes of CFW annually, the highest among South Asian countries (Hamish Forbes and Tom Quested, 2021). This humongous amount of CFW is not appropriately managed, most of it gets disposed into landfills that lead to various ecological issues. Valorization of CFW into value-added fuels such as bio-oils through thermochemical conversion processes can lessen the landfill's ecological impact and provides an alternative to the global dependency on crude reserves (Bayat et al., 2021). Slow pyrolysis is a prominently used thermochemical process to produce bio-oil from lignocellulosic biomass by depolymerizing three-building biopolymers such as cellulose hemicellulose and lignin under the action of heat in an inert atmosphere. In comparison with other thermal processes such as hydrothermal liquefaction, bio-oil production through slow pyrolysis is less energy intensive (Bayat et al., 2021; Mullen et al., 2009). Moreover, bio-oil from slow pyrolysis has the potential to be mixed with other transportation fuels after catalytic upgradation.

Bio-oil from slow pyrolysis is a dark brown liquid that contains a wide range of complex organic compounds, including carboxylic acids, ketones, esters, aldehydes, phenolics, alcohols, water. Hence, the characterization of bio-oil is an essential element of pyrolysis research since it lays the groundwork for future biofuel development, including bio-oil upgradation and process improvement. In addition, the composition of bio-oil is affected by several factors, including heating rates, pyrolysis temperatures, residence time, and the quality of lignocellulosic biomass used as a pyrolysis feedstock (Mullen et al., 2009; Negahdar et al., 2016). Researchers have worked on producing pyrolysis bio-oil from various wastes and evaluated the bio-oil characteristics. Slow pyrolysis of linseed residue produced roughly 42% bio-oil, which was further examined using NMR and FTIR techniques, mainly revealing the aliphatic and aromatic groups (Acikgoz and Kockar, 2009). Similarly, slow pyrolysis of *Prunus Armeniaca* seed residue yielded around 45%

* Corresponding author.
E-mail address: priyanka@snu.edu.in (P. Katiyar).

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bio-oil, and column chromatography majorly showed the presence of polar compounds as well as aliphatic compounds, which are methanol and n-hexane soluble, respectively (Fadhil, 2017).

Furthermore, *Pomgamia* de-oiled cake was employed as a feedstock in a slow pyrolysis process and produced bio-oil yield of roughly 30%, with the highest percentage of short aliphatics as well as olefinic groups as highlighted in ^{13}C and ^{1}H NMR analysis (Chutia et al., 2014). Authors have also studied the slow pyrolysis of sawdust in the batch process and found the bio-oil yield around (~45%). The ^{1}H NMR and ^{13}C NMR of saw dust bio-oil confirmed the presence of alkyl, and aromatic groups in the bio-oil composition (Soni and Karmee, 2020). Researchers have worked on various food waste, for example, Qing et al. (2022) performed the pyrolysis of food waste collected from a food waste treatment plant in China with varying temperature (400–700 °C) and concluded that bio-oil yield increases with temperature where N-compounds contributes 40% (highest) of total bio-oil at 500 °C as per GC–MS analysis (Qing et al., 2022). In another study on food waste, discarded vegetables and fruits (banana, orange, onion, bell pepper peels etc.) were taken as feedstock for the pyrolysis at varying temperature, reaction time and heating rate to maximize the biochar yield (Patra et al., 2021). Maximum biochar and bio-oil yield was found at 300 °C and 500 °C respectively and the bio-oil was enriched with acids, alcohols and phenols (Patra et al., 2021). Co-pyrolysis of food waste (kitchen waste) was performed with low-density polyethylene which resulted a 20% increase in bio-oil yield (Neha and Remya, 2021). Therefore, it can be noticed that the bio-oil composition from slow pyrolysis differs significantly with the change in the type of biomass or waste. It is also evident that scarce studies are available on the generation and analysis of bio-oil obtained from food waste however no study reports the generation and comprehensive characterization of bio-oil produced from the cooked food waste (collected from restaurants, dining halls etc) which is a main concern of disposal now a days. The cooked food waste considered in this study is quite different from typical biomass and other lignocellulosic waste in terms of its chemical composition. This type of waste may contain additional compounds such as starch, protein, waxes, fats, and oils in significant quantity and may produce bio-oil with distinctively different composition (Singh and Yadav, 2019; Yadav et al., 2021). However published literature lacks such in-depth research on cooked food waste that may, one hand, give a solution for waste disposal and, on the other hand, provide an appropriate feedstock for green fuel and chemicals generation to cater the depletion in fossil fuel.

Therefore in present study, mixed cooked food waste was considered as feedstock to produce bio-oil using slow pyrolysis process by varying the pyrolysis temperature (300–500 °C). Physico-chemical properties of cooked food waste were evaluated using proximate, elemental, lignocellulosic, and nutrient analysis. A comparison between bio-oil yield at different temperature was made to predict optimized pyrolysis temperature to maximize the bio-oil yield. Further to study the effect of pyrolysis temperature on the composition of bio-oil, sophisticated characterization techniques such as ^{1}H and ^{13}C NMR (Nuclear magnetic resonance) and GC–MS have been employed to understand the complex nature of bio-oil obtained. NMR (Nuclear magnetic resonance) technique can examine the entire portion of bio-oil, by giving a detailed, comprehensive analysis of the structure as well as the orientation of the organic compounds present in the bio-oil (Wang et al., 2020). The ^{1}H and ^{13}C NMR analysis not only provide the structural information but also depicts the appropriate ratio of protons and carbon atoms present in the bio-oil sample, i.e., aliphatic to aromatic ratio. Though, ^{1}H NMR predicts the whole proton content of the Bio-oil but provides little information about the complete distinction between different functional groups present in the Bio-oil. On the other hand, ^{13}C NMR provides more accurate information about functional groups with less spectral overlap and a wider chemical shift range. The ^{13}C NMR can successfully establish the conversion of carbohydrates and other bio-biopolymers in Ligno-cellulosic biomass by suggesting tentative mechanisms involved in the thermochemical conversion process to its products. The DEPT (Distortionless Enhancement by Polarization Transfer) and NMR analysis give an in-depth idea about the different types of carbons, such as primary, secondary, tertiary, and quaternary present in the bio-oil. Gas chromatography (GC) further helps to confirm the presence of various compounds and quantify them in their respective proportion. Therefore, NMR is used as a qualitative means to characterize the bio-oil to analyze its nature (aliphatic or aromatic). Later, the compounds are confirmed and quantified using the GC–MS Analysis technique.

2. Materials and methods

2.1. Sample preparation

Cooked Food waste comprising of waste vegetarian and non-vegetarian food was collected from the Dining Halls of Shiv Nadar University. It was sun-dried for one week after collection and then milled and sieved to produce particles of various sizes. The particle size used in this study was in the range of 10–18 mm in diameter.

2.2. Physico-chemical characterization of CFW

The Physico-chemical characterisation of CFW was conducted using proximate, elemental, lignocellulosic, and nutrient analysis. The proximate analysis deals with the quantitative estimation of moisture, volatile matter, ash, and fixed carbon content. Their content was measured in accordance with ASTM D-3173-11 standard, ASTM D 3175-11 standard, ASTM D-3174-11 standard, respectively. The Fixed carbon is calculated using the following Equation (1) (Singh and Yadav, 2019).

$$\text{Fixed Carbon } (\backslash\%) = \ 100\ \text{-}) = 100 - [\text{Water content}\,(\%) + \text{Volatile Matter}\,(\%) + \text{Ash}\,(\%)]$$

Elemental analysis was performed to estimate the composition of elements (C, H, N, S, O) present in the CFW sample. Elemental analysis was done using Thermo Finnigan Flash 1112 Series Elemental analyser (Singh and Yadav, 2019). Additionally, the lignocellulosic analysis was conducted using Tappi T222 om-02 standard method for lignin content, TM I – A11 2001 method for hemicellulose, and TM1 – A9 2001 method for holo-cellulose content. Cellulose content was determined by subtracting hemicellulose from hollo-cellulose content. Furthermore, nutrient analysis was conducted to get the starch, protein, and lipids content in the food waste sample. AOAC (2001.11) Official Method was followed for quantitative estimation of crude protein, also known as Kjeldahl method. The starch content was determined using two methods: i) Anthrone Method and ii) Phenol-sulphuric acid method. The lipids were quantified using (AOAC 2003.05) protocol with the help of organic solvent such as n-hexane in the Soxhlet extraction apparatus.

2.3. Pyrolysis reactor set-up and procedure

The experimental setup used in this study is shown in Fig. 1. It's a down-draft fixed bed reactor, fitted with a hopper and N_2 preheater at the top along with a cold water condenser and a 2-neck round bottom flask at the bottom. The side neck was used to vent the gases into the

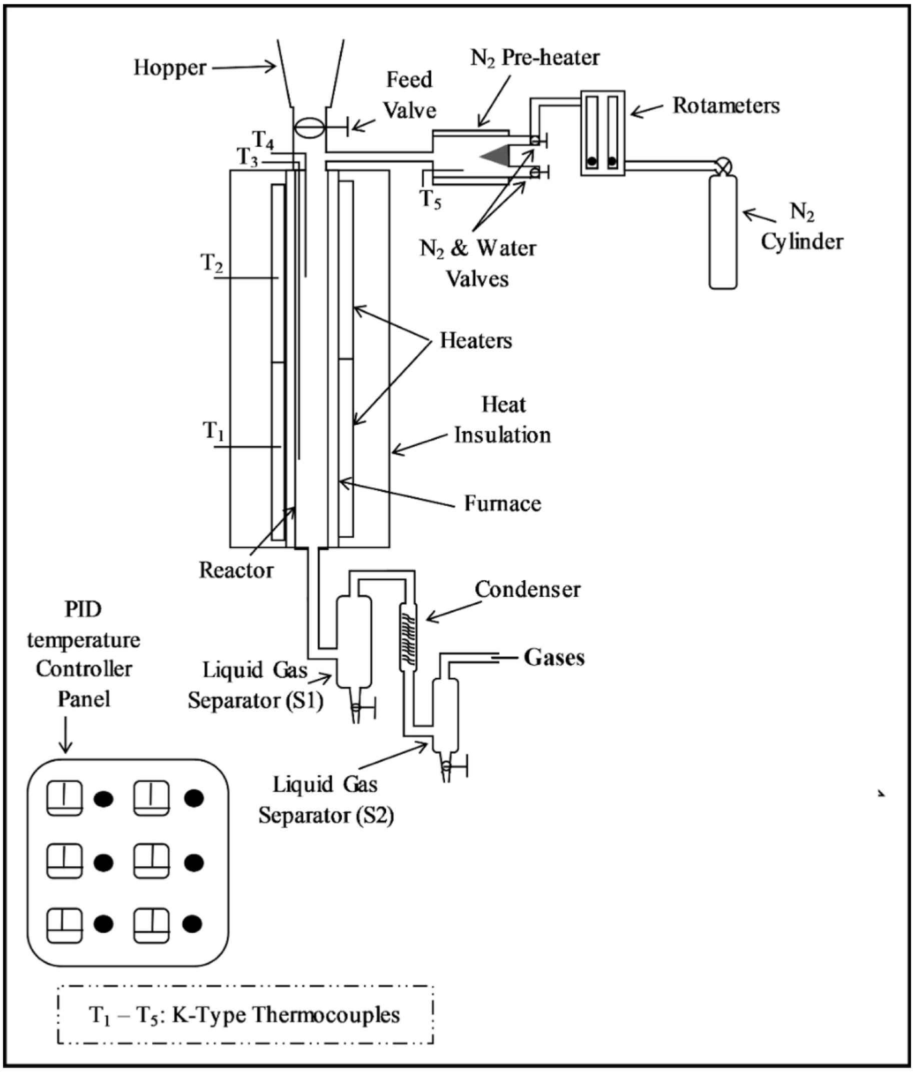


Fig. 1. Pyrolysis Reactor Setup.

atmosphere. 100 gm of sun-dried CFW was charged through a hopper and nitrogen was fed at a flow rate of 100 mL/min to facilitate the pyrolysis effectively. Nitrogen also helps to maintain the inert environment inside the reactor.

The heating of the reactor was started once the CFW was fed to reactor at the heating rate of 10 °C/min. As the reactor reached the required temperature (300–500 °C), it was maintained there for 1 h for pyrolysis. A PID controller was used to regulate the heater's temperature. During the whole pyrolysis process, the condensed bio-oil was collected in the bottom flask, which was then purified and characterized.

2.4. Sample purification

Bio-oil obtained from the pyrolysis needed to be purified using an appropriate solvent and dehydrating agent prior to characterization to get noiseless signals. Bio-oil samples were purified using Ethyl Acetate to solubilize both polar and non-polar groups in Bio-oil. Further, solvent-extracted bio-oil was treated with anhydrous Na_2SO_4 to remove residual moisture and then filtered in 40 μm filter paper to remove any suspended particle from the solvent-extracted bio-oil fraction.

2.5. ^1H NMR, ^{13}C NMR & DEPT analysis of bio-oils from CFW.

The NMR analysis of Bio-oil was done to investigate its overall molecular identity formed during pyrolysis. The ^1H NMR and ^{13}C NMR analyses were done using Bruker 400 MHz NMR Machine. A 200 μl of bio-oil sample was mixed in DMSO-d_6 and then subjected to ^1H NMR analysis and ^{13}C NMR analysis. The ^1H NMR was recorded at 400 MHz and 23 °C, whereas ^{13}C NMR was measured at 100 MHz and 23 °C. The scanning ranges for ^1H NMR and ^{13}C NMR analysis are 0–10 ppm and 0–225 ppm, respectively. Topspin software (Bruker, Germany) was used to analyze the signals and then normalize them to the overall area for visualization and quantification of functional groups. The area of the solvent peak (DMSO-d_6) was excluded for area integration. Additionally, the DEPT analysis was done to identity the distribution of CH, CH, CH_3 groups which are arising from the organic components present in the bio-oils. The selection angle parameter was set at 90 and 135 degrees for DEPT 90 and 135, respectively. The DEPT 90 spectra show upwards facing peaks for CH Groups, whereas the DEPT 135 spectra show upwards facing peaks for CH_3 and CH groups and downwards facing peaks for CH_2 groups.

2.6. GC–MS analysis of Bio-oils from CFW

GC–MS analysis was carried out in Agilent 5977B GC/MSD where HP-5MS capillary column (30 m × 0.25 mm, 0.25 μm) was used. The purified bio-oils were dissolved in chloroform (HPLC grade) and injected at a split ratio of 1:20. The ramping rate was maintained at 2 °C/ min.

Table 1
Physico-chemical properties of CFW.

Proximate Analysis			
Moisture (%)	**Volatile Matter (%)**	**Ash (%)**	**Fixed Carbon (%)**
7.14	75.28	2.55	15.02
Ultimate Analysis			
Carbon (%)	**Hydrogen (%)**	**Oxygen (%)**	**Nitrogen (%)**
46.2	6.5	40.4	1.94
Lignocellulosic components			
Lignin (%)	**Cellulose (%)**	**Hemi-cellulose (%)**	
15.04	26.20	5.05	
Nutrients in CFW			
Lipids (%)	**Starch (%)**	**Crude Protein (%)**	
25.02	14.13	12.15	
% Bio-Oil Yield (g/g)			
BO-300	**BO-400**	**BO-500**	
31%	45%	41%	

The initial and final temperatures were maintained at 60 °C and 300 °C, respectively. Helium was used as a carrier gas at a flow rate of 3 mL/min. After the peaks were obtained, organic compounds matching respectable peaks were identified using NIST-MS search software.

3. Results and discussions

3.1. Physico-chemical analysis of CFW

The proximate analysis given in Table 1 shows the proportion of moisture, volatile matter, ash, fixed carbon content, and HHV value of CFW. The CFW has ~75% volatile matter that was close to volatile matters present in typical biomass and fixed carbon (~15%) was less as compared to its content in typical biomass (>20%)(Singh and Yadav, 2019).

Table 1 also shows the elemental composition of the CFW sample obtained from the ultimate analysis. The carbon and hydrogen content was comparable with their content in any other biomass (Adamovics et al., 2018). More hydrogen per carbon would produce more energy during the oxidation reaction and is generally characterised by the H/C ratio. Additionally, oxygen was present in a higher amount than that in woody biomass and could contribute towards the more combustibility or oxidation of the CFW, an undesirable property in biomass (Adamovics et al., 2018). Furthermore, there were few traces of nitrogen in the CFW sample, which could produce harmful NOx during thermochemical treatment (Mansor et al., 2018).

Table 1 shows the percent composition of cellulose, hemicellulose, and lignin, along with the composition of various nutrients in the CFW sample. The combined percent composition of ligno-cellulosic compounds was found to be around 49%, the remaining 51% consisted of moisture (7.14%), ash (2.55%), and various other nutrients (41.3%). The individual proportion of ligno-cellulosic compounds in the CFW sample was 5.03% for hemicellulose, 26.20% for cellulose, and 13.04% for lignin. The percent composition of hemicellulose in the CFW was quite less because it got degraded significantly during the cooking of the food as it was an easily degradable compound at even low temperatures (≤200 ^{0}C). Consecutively, its composition fell below the recommended range of hemicellulose percent composition in any biomass (20–35%) (Wei et al., 2017). However, the percent composition of cellulose and lignin was within the range of their percent composition in typical biomass (25–50% for cellulose and 10–25% for lignin) (Wei et al., 2017).

Table 1 also shows presence of additional compounds in the form of protein (12.15%), starch (14.13%), 25.02% lipids (25.02%). Since, these compounds are present in very large fraction (>40%), they can significantly affect the characteristics of bio-oil generated from slow pyrolysis of CFW as compared to the bio-oil characteristics obtained from the slow pyrolysis of any other waste or biomass.

3.2. Effect of pyrolysis temperature on the bio-oil yield

The bio-oil yield from slow pyrolysis at different temperatures is tabulated in Table 1. At 300 °C, CFW has undergone torrefaction or early-stage pyrolysis, where one of the major products was biochar, not the bio-oil, therefore, bio-yield was less at this temperature. Slow pyrolysis of CFW at 400 °C produced bio-oil with the highest yield (~45%) as compared to the bio-oil yield at the other two temperatures.

The bio-oil is generated majorly due to the decomposition and depolymerization of various carbohydrates and proteins present in CFW at this temperature and from the melting of oils and waxes in CFW. Other simultaneous reactions were involved in maximizing the bio-oil yield such as cyclization, aromatization, and rearrangement of ligno-cellulosic components present in CFW to form liquid components which are ultimately obtained in the bio-oil (Hu and Gholizadeh, 2019). Furthermore, on increasing the temperature to 500 °C, the bio-oil yield from slow pyrolysis of CFW decreased due to the generation of higher syngas fraction in pyrolysis products. The secondary cracking of vapors obtained during pyrolysis led to the formation of non -condensable vapors such as CO_2, CO, H_2, N_2, O_2, and CH_4 (Kumari and Mohanty, 2020).

3.3. ^{1}H NMR analysis of Bio-oil obtained at various temperatures

^{1}H NMR spectrum for bio-oil obtained from CFW pyrolysis at different temperatures is presented as BO-300, BO-400, and BO-500, respectively in Fig. 2 (a-c). It shows the crowding of multiple peaks with varying intensities from (0.5–9 ppm), indicating the presence of various compounds and multiple functional groups in the bio-oil. These NMR spectra of bio-oil from CFW are comparable with NMR analysis of bio-oil obtained from slow pyrolysis of linseed (Acikgoz and Kockar, 2009), Pongamia deoiled cake (Chutia et al., 2014), sawdust (Soni and Karmee, 2020) and apricot kernel shell (Demiral and Kul, 2014). The ^{1}H NMR characterization of linseed pyrolysis bio-oil showed the significant accumulation of peaks with varying intensity in the region (1–2.5 ppm) and a singlet in (6–8 ppm) which indicates the significant presence of aliphatic compounds and less aromatic compounds, respectively. The same kind of ^{1}H NMR characteristics was also shown by bio-oil obtained from apricot, where a majority of crowding was demonstrated in the aliphatic region, a singlet was observed in the methoxy region, and few peaks were observed in the aromatic region. However, in the ^{1}H NMR characteristics of bio-oil from sawdust the methoxy region was equally crowded as of aliphatics region. In the present work, the majority of the crowding can be seen from 0 to 3 ppm, due to the presence of aliphatics and alkanes in bio-oils obtained from CFW pyrolysis at different temperatures.

In the NMR of BO-300, peaks with lower intensities can be seen in the first (0–1.6 ppm) chemical shift area, indicating the presence of CH_3, CH_2, and alkanes, whereas in BO-400 and BO-500, the crowding of peaks has reduced.However, it shows the presence of singlets with higher intensities, in the bio-oil samples BO-400 and BO-500 indicating the presence of more alkanes in the bio-oil at higher temperatures. Additionally, in the NMR of BO-500, the intensity of singlets in the first region (0–1.6 ppm) was higher than that of BO-400 and BO-300, indicating the presence of more alkenes in the bio-oil from slow pyrolysis of CFW at a higher temperature.

Furthermore, the region of 1.6–2.2 in NMR of BO-300 shows many peaks with varying intensities. Singlet at 1.8 ppm has the highest intensity and remains constant at all temperatures. These peaks indicate the presence of methylene group proton and aliphatic hydroxyl group proton. However, with temperature, the intensities of other peaks varied and were found to decrease for BO-400 NMR and then increase again for

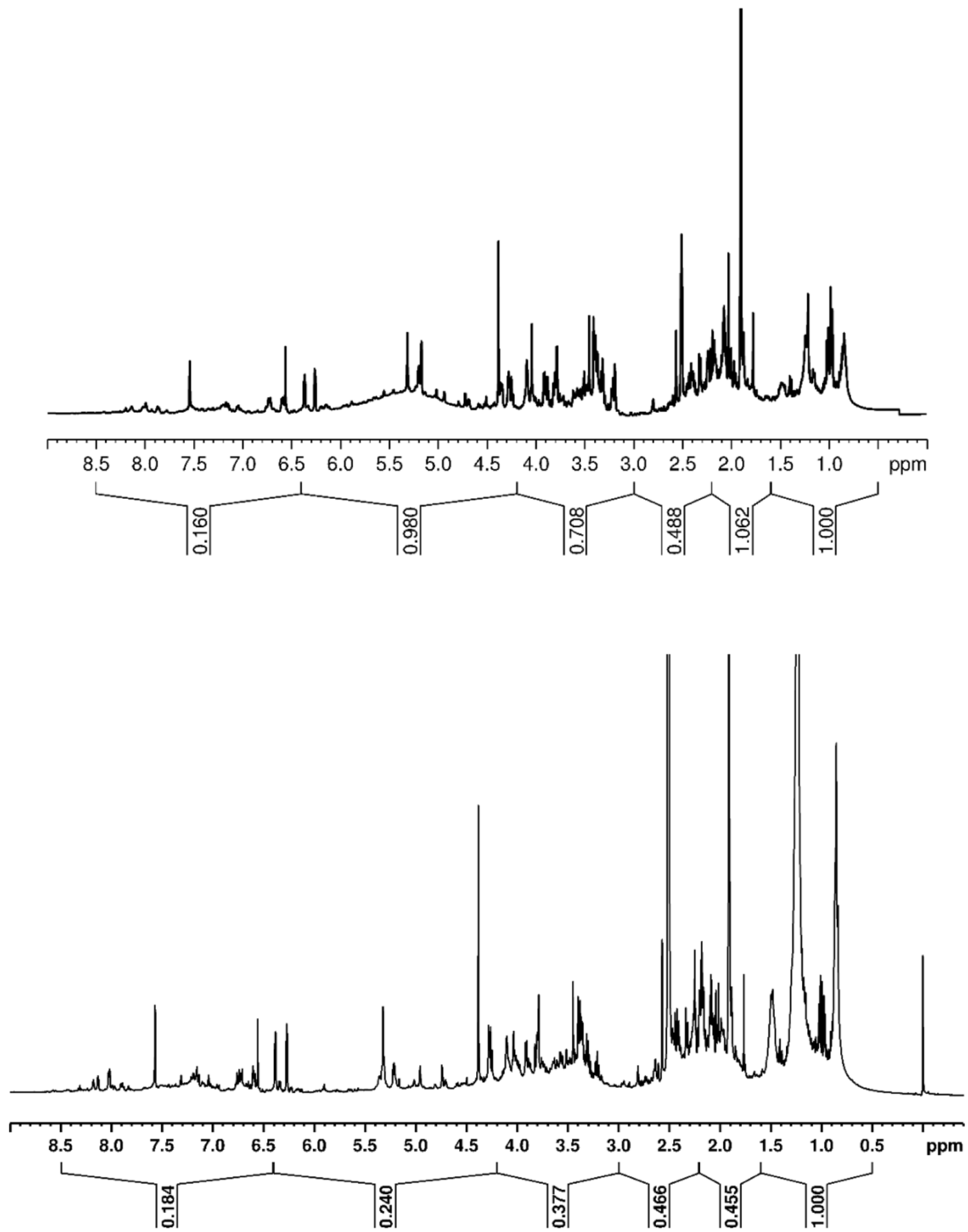


Fig. 2. (a)^{1}H NMR of BO-300. Fig. 2b): ^{1}H NMR of BO-400. Fig. 2(c): ^{1}H NMR of BO-500.

BO-500, thus indicating a slight reduction and then a small increment of these groups in BO-400 and BO-500, respectively. The exact variation in methylene and hydroxyl groups has been determined by using the area integration method and presented in Table 2.

The next region (2.2–3.0 ppm) shows the presence of one solvent singlet peak (DMSO-d_6) at 2.5 ppm in all bio-oils (BO-300, BO-400, and BO-500). Apart from this solvent peak, there are other peaks with varying intensities in this region. These peaks indicate the protons on carbons directly bonded to an aromatic ring, called benzylic protons in this region and their variation with temperature. Additionally, NMR of the region, 3.0–4.2 ppm, shows that BO-300 and BO-400 exhibited many peaks of varying intensities, indicating the presence of alcoholic group protons, which are less affected when the pyrolysis temperature increases from 300 °C to 400 °C. However, when the pyrolysis temperature increased further to 500 °C, the intensity and the crowding of peaks have drastically reduced, indicating the reduction in the –CH_2O, CH_3O, and alcoholic protons in the bio-oil. The same has been determined by the area integration method and reported in Table 2.

Furthermore, in the 4–6.4 ppm region, BO-300 exhibited a broad hump with 3 prominent singlets and micro peaks; however, for BO-400, the hump and the intensity of the 2 singlets, reduced except for the singlet at 4.4 ppm. This indicated the reduction in the presence of methoxy and carbohydrate protons in the bio-oil as the temperature increased from 300 °C to 400 °C. However, in the NMR of BO-500 the peaks have considerably reduced in this region, indicating the small presence of methoxy as well as carbohydrate protons. In the 6.4–8.5 ppm region, BO-300 showed the presence of peaks with 2 singlets at 6.6 ppm & 7.6 ppm, indicating small presence of heteroaromatic or aryl protons (Hao et al., 2016; Ingram et al., 2008). However, for BO-400, the intensity of the singlet at 7.6 ppm increased, indicating the slight increase in the presence of heteroaromatic or aryl protons. For BO-500, there was only one singlet at 7.6 ppm, but micro peaks further

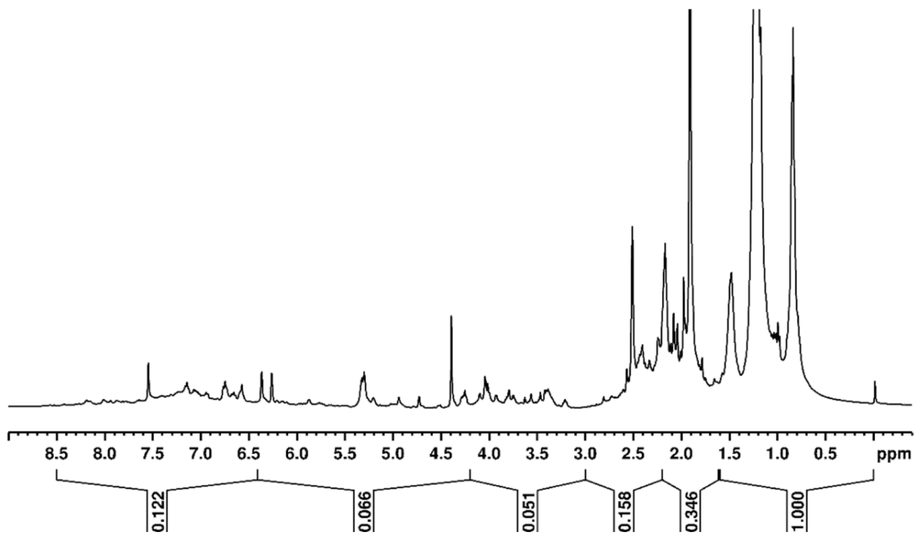


Fig. 2. (*continued*).

Table 2
[1]H NMR and [13]C NMR analysis of CFW-Bio-oil using the area integration.

H-NMR Analysis of CFW-Bio-oils		% Area		
Region	Proton Assignment	BO-300	BO-400	BO-500
0–1.6	CH_3, CH_2 and Alkanes	23.33	39.96	58.52
1.6–2.2	CH_2, Aliphatic, –OH	24.78	18.17	20.27
2.2–3.0	CH_3C=O, CH_3-Ar, –CH_2-Ar or	8.77	9.85	7.19
3.0–4.2	CH_3O, –CH_2O, Alcohols, and probable presence of Methylene dibenzene	16.51	15.04	2.99
4.2–6.4	CHO, ArOH, HC=C, Methoxy groups or carbohydrates	22.87	9.59	3.87
6.4–8.5	ArH, HC=C (Conjugated), Heteroaromatics	3.72	7.36	7.14

[13]C NMR Analysis of CFW-Bio-oils		% Area		
Region	Carbon Assignment	BO-300	BO-400	BO-500
0–28	Short Aliphatics	26.0	26.24	30.47
28–55	Long and branched aliphatics (excluding solvent peak (dmso-d6)	16.0	23.63	34.18
55–95	Alcohols,ethers,phenolic-methoxys, carbohydrates sugars	30.9	19.68	7.28
95–165	Aromatics, Olefins	20.5	20.04	19.24
165–180	Esters, Carboxylic acids	5.37	4.49	4.65
180–215	Ketones, aldehydes	1.13	5.89	4.16

broadened in this region (-6.5–7.4) ppm, suggesting a small change in the presence of heteroaromatic or aryl protons.

3.4. [13]C NMR analysis of bio-oil obtained at various temperatures

[13]C NMR spectrum for BO-300, BO-400, and BO-500 are shown in Fig. 3(a-c) and indicates the presence of various compounds in the bio-oil obtained from slow pyrolysis of food waste (Alwehaibi et al., 2016). It can be noticed that there is crowding of peaks in multiple regions from 10 to 215 ppm, which indicates the presence of aliphatics, alcohols, carbohydrates, aromatics, and esters in the bio-oil. The same has been reported in Table 2.

In the 28–55 ppm region, there is a presence of dmso-d6 solvent singlet at 40 ppm in the NMR of all bio-oil samples (BO-300, BO-400 & BO-500). Apart from a solvent singlet, there are other peaks with varying intensities in the same region for all the bio-oil samples. It can be perceived that the intensities of peaks increased with increasing pyrolysis temperature due to an increase in carbons of long and branched aliphatic in bio-oil. In the next region (55–95 ppm), multiple clustered singlets are present in all bio-oil samples, indicating the presence of alcohols, ethers, phenolic-methoxy, and carbohydrate sugars. However, their presence in the bio-oil decreases with increased pyrolysis temperature as indicated by decreasing peaks of the clustered singlets. The same has been reported in Table 2.

Furthermore, a region between 95 and 165 ppm exhibits scattered peaks due to aromatic compounds whose intensity first increased in the [13]C NMR of bio-oil from pyrolysis at 400 °C and then decreased a bit for 500 °C. The presence of aromatic components are favourable for bio-oil synthetic modification (Khuenkaeo et al., 2021). The NMR region between 165 and 180 ppm for BO300 shows a singlet along with 2 micro peaks due to the presence of esters and carboxyl's, however, no change in peak intensity is observed with an increase in pyrolysis temperature from 300 °C to 400 °C. The [13]C NMR spectra in the region 180–215 ppm indicate the presence of ketones and aldehyde groups in bio-oil. It can be noticed that there is no significant peak present in this reason, suggesting very less presence of these groups in the bio-oil.

3.5. GC–MS analysis of Bio-oils obtained at various pyrolysis temperatures

The fractions of organic components obtained from the GC–MS analysis of BO-300, BO-400 & BO-500 are provided in the Table 3. It can be noticed that many compounds were present in BO-300 however not present in BO-400 & BO-500 and vice-versa. In BO-300, total 41 compounds were out of which 2 compounds (Furanmethanol and Levoglucosan) were present in large quantity (>15%), 7 compounds were present in moderate amount (>3% fraction) such as 2-cyclopenten-1-one,3-methyl, Butyrolactone,2-cyclopentanedione,3-methyl etc. which are mentioned in Table 3. The remaining compounds were present in less quantity (<3% fraction) such Ethanone, Phenol, Hexanoic acid, Catechol etc. Similarly, in BO-400 & BO-500, total 44 and 28

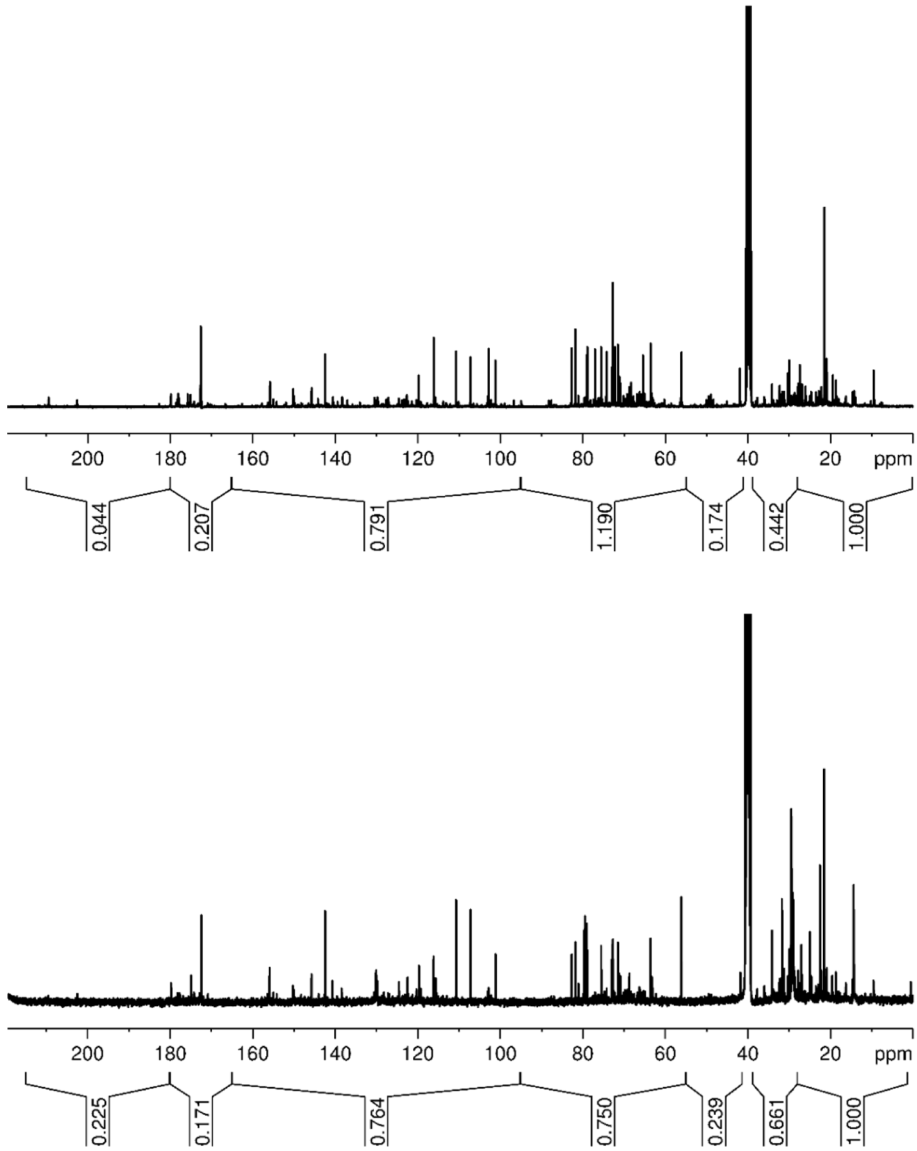


Fig. 3. (a). 13-C NMR analysis of BO-300. Fig. 3b. ^{13}C NMR analysis of BO-400. Fig. 3(c): ^{13}C NMR analysis of BO-500.

compounds were detected, respectively. In BO-400, 3 compounds were present in large fraction (>10%), whereas in BO-500, only one compound was present in a very large fraction (>50%). Furthermore, in BO-400, 2 compounds were present in a moderate fraction, and the remaining 36 compounds were present in a small fraction (<3%). In contrast, in BO-500, only one compound was present in a moderate fraction (>3%) and remaining 26 were present in a small fraction.

There were only 5 compounds that were present in all the bio-oil samples, including *2*-Furanmethanol, 1,6-Anhydro-alpha-d-galactofuranose (Levoglucosan), Phenylacetamide,4-fluoro-N-butyl, Ethanone,1-(2-furanyl)-, and Phenol. There were two other compounds (Succinimide, Catechol) that were present in both the BO-300 and BO-400 samples and 10 other compounds that were present in both BO-400 and BO-500 samples, such as Benzene,1,3-bis(3-phenoxyphenoxy)-, p-Cresol, n-Hexadecanoic acid, 2,4-Di-*tert*-butyl-phenol, 2-Cyclopenten-1-one, 2-Hydroxy-3-Methyl-, Cetene, Benzyl Alcohol, Octanoic Acid, 3-cyclopentene-1-acetaldehyde,2-oxo-, and Hexadecane.

It can be noticed from Table 3 that the fraction of 2-Furanmethanol in BO-400 went down to ~15% from 26% in BO-300. Similarly, a fraction of the β form of levoglucosans increased significantly, suggesting the conversion of α- levoglucosans to β- glucosans at a higher pyrolysis temperature. Levoglucosans are used in producing bioethanol by hydrolysis reaction (Anand Ramanathan et al., 2022). It can also be noticed that the presence of levoglucosans reduced significantly to less than 3% in bio-oil (BO-500) and 2-Furanmethanol reduced to around 8% in BO-500. Additionally, the fraction of aromatics (Benzene, 1,3-bis (3-phenoxyphenoxy))- increased manifold (to >60%) in BO-500 as compared to an aromatic fraction in other two bio-oil samples (BO-300 & BO-400). % of aromatic compounds present in bio-oil obtained from cooked food waste is lesser than that of other food waste such as palm kernel shell (81%) (Wan Mahari et al., 2022) however way more than aromatic content found in bio oil obtained from pyrolysis of other types of waste (Wan Mahari et al., 2022). Presence of more aromatic compounds in bio-oil leads to higher stability of the bio-oil. Additionally, many of these aromatic compounds in bio-oil can be used as feedstock for synthesizing many chemicals (Shivhare et al., 2021).

Furthermore depolymerization of lignin is greatly enhanced at 350 °C which facilitates the formation of smaller molecules however it can be seen that further increase in temperature up to 500 °C resulting in repolymerization of several small molecules and forming long chain

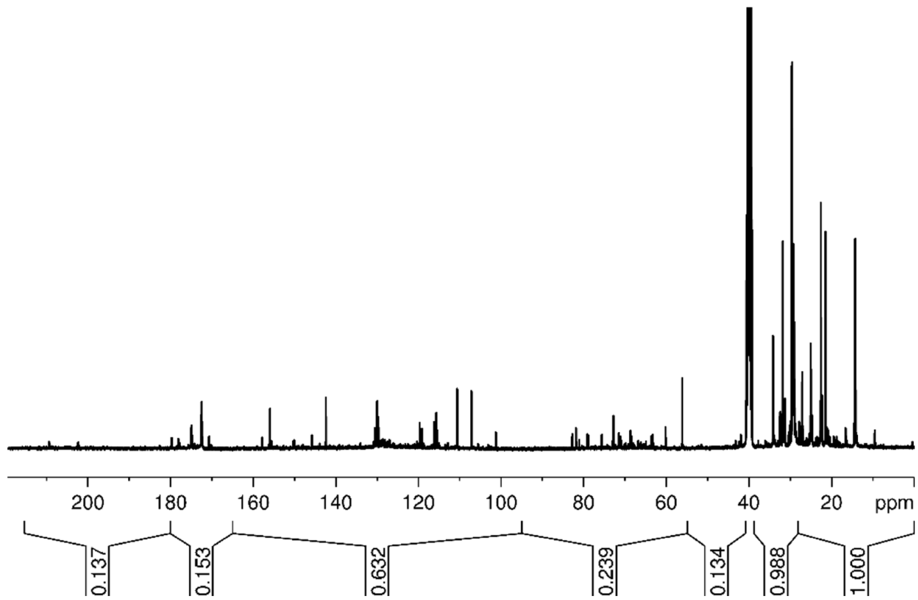


Fig. 3. (*continued*).

aromatic ether i.e. Benzene, 1,3-bis (3-phenoxyphenoxy) (Huang et al., 2017). To enhance the cleavage of aromatic ether bonds present in lignin and avoid the repolymerization of small molecules, use of catalysts such as metal oxide-based catalysts, bimetallic catalyst and Ru/C etc. are recommended (Wu et al., 2018). Furthermore, the presence of different cyclopentene and cyclopentane also reduced in BO-400. However, the fraction of phenols was much more in BO-400 as compared to its fraction in BO-300.

The decomposition of lignin content of biomass is responsible for presence of phenols in bio-oil which starts degrading after 350 °C and below 500 °C (Chen and Kuo, 2011; Zhao et al., 2021). Lignin, a phenolic polymer, protects the cell wall polysaccharides from being digested by enzymes and makes for a tough, long-lasting composite material which is degradable at high temperature (Chen and Kuo, 2011). However, an elevated temperature results in the improved degradation of the obtained phenols which causes a decrease in the phenol content in bio-oil from pyrolysis at higher temperature (Zhao et al., 2021). A similar variation can be seen in other compounds between BO-300, BO-400, and BO-500. Furthermore few organic nitrogen compounds in small amount (<3%) were also found in food waste bio-oil such as, Pyridine, Proline, Hexadecanenitrile, Octanenitrile and Benzylamine as mentioned in Table 3 which mainly come from the pyrolysis of protein content of CFW.

3.6. DEPT 135 and 90 of BO-300

DEPT 135 and 90 spectra for BO-300,400 and 500 are shown in Fig. 4. S1-S6 of supplementary material, respectively. However, DEPT confirmation of the compounds identified by GC–MS for BO-300,400 and 500 are presented in Table 4.

3.7. Mechanistic analysis of variation in bio-oil composition

Fig. 4(a) and (b) show the changes in the fraction of various organic groups of compounds and the reaction mechanism of pyrolysis of CFW respectively. The content of carbonyl, phenolic or alcoholic, aromatics, aliphatics, anhydrous sugars, furans, nitrogen compounds, and acids in bio-oil vary with changes in temperatures. The carbonyl compounds (Aldehydes, Ketones, Oxo groups, and Esters) were significantly present in BO-300, however, their fraction decreased in the bio-oil with an increase in pyrolysis temperature from 300 to 500 °C. This trend can be attributed to the thermal degradation of lipids or long-chain triglycerides and starches at higher temperatures, leading to the formation of smaller carbonyl compounds (Chagas et al., 2016). Additionally, hemicellulose and part of cellulose start to depolymerize at temperatures beyond 240 °C, giving rise to significant carbonyl compounds at BO-300 (Wang et al., 2013). As the pyrolysis temperature is increased, carbonyl compounds get oxidized into carboxylic acid compounds, alcohols, and phenolic compounds. After 400 °C, the residual carboxyl groups may begin fragmenting, rearrangement, cyclization, and aromatization, further diminishing their fractions in bio oils produced at 400 °C and 500 °C as can be noticed in Fig. 4(a).

The fraction of phenolic and alcoholic compounds was less in BO-300, however, it increased as the pyrolysis temperature increased from 300 °C to 400 °C, and then decreased on further increase in pyrolysis temperature to 500 °C. This happened because at low temperatures (300 °C), only hemicellulose would decompose and produce phenols and alcohols, however, it was not present much in the food waste sample as it had already degraded significantly during cooking (Jiang et al., 2010). The increase in pyrolysis temperature to 400 °C led to lignin degradation into phenols, and conversion of some carbonyl compounds into alcohols and phenols, thus increasing the alcoholic and phenolic fraction in BO-400. Further increase in pyrolysis temperature to 500 °C caused the breaking down of phenolic groups, leading to the formation of complex aromatic compounds. Catechols which were present in BO-400, were absent in BO-500 because these get started to gasify at this temperature, thus reducing the phenolic and alcoholic content in BO-500 (Mohan et al., 2017). Similarly, aliphatic compounds and furans were converted into aromatics at high temperatures due to cyclizing and aromatization to increase the aromatics fraction sharply at 500 °C (Khan et al., 2019; Wang et al., 2021). The fraction of aromatics in bio-oil from pyrolysis at high temperature (~500 C) was compared with the values reported from other biomass (Acikgoz and Kockar, 2009), and it was found that the aromatic fraction was much higher (>60%) in this case.

Furthermore, Anhydro sugars or Levoglucosan content in the bio-oil BO-300 was high because the cellulose molecule started getting activated and formed active cellulose at temperatures between 150 °C and 290 °C, which in turn converted into levoglucosan at 300 °C (McGrath et al., 2003). On further increase in pyrolysis temperature, levoglucosan

Table 3
GC–MS Analysis of BO-300, BO-400 and BO-500.

S. No.	Retention Time	Compound Name (Detected from NIST GCMS software)	% Peak Area		
			BO-300	BO-400	BO-500
1	4.42	2-Furanmethanol	25.97	15.18	8.51
2	17.67	1,6-Anhydro-alpha-d-galactofuranose (Levoglucosan)	16.65	14.27	2.81
3	26.90	1,6-Anhydro-beta-d-talopyeanose (Levoglucosan)	4.88	0.22	–
4	7.63	2-cyclopenten-1-one,3-methyl-	4.58	–	–
5	4.05	2-cyclopenten-1-one	4.28	–	–
6	5.96	Butyrolactone	4.17	–	–
7	9.82	1,2-cyclopentanedione,3-methyl-	4.09	–	–
8	7.52	Phenylacetamide,4-fluoro-N-butly	3.70	1.86	0.98
9	11.55	Furan,3-methyl-	3.27	–	–
10	5.89	Ethanone,1-(2-furanyl)-	2.27	1.62	1.16
11	11.87	3-octen-1-ol,(E) -	2.20	–	–
12	3.11	Oxirane,2-methyl-3-(1-methylethyl)	2.05	–	–
13	8.10	Phenol	1.85	3.33	1.64
14	18.49	3,4-Anhydro-d-galactosan	1.60	–	–
15	4.89	1,6-Anhydro-beta-d-talopyeanose (Levoglucosan)	1.45	–	–
16	7.95	Hexanoic acid	1.32	–	–
17	5.77	2-cyclopenten-1-one,2 methyl	1.30	–	–
18	3.82	Pyridine,2-methyl	1.30	–	–
19	20.54	3(2H)-Pyridasinone,6-methyl	1.16	–	–
20	10.38	Furan,2,3,5-trimethyl-	1.05	–	–
21	16.69	Valeric anhydride	1.02	–	–
22	4.76	2-Propanone,1-(acetyloxy)-	0.90	–	–
23	4.14	3-Methylheptyl acetate	0.88	–	–
24	14.02	Succinimide	0.80	1.78	–
25	8.80	Di-n-decylsulfone	0.75	–	–
26	20.87	Dianhydromannitol	0.61	–	–
27	18.21	2-Undecene,4,5-dimethyl-	0.61	–	–
28	11.04	Furan,tetrahydro-2,4-dimethyl-,cis-	0.57	–	–
29	28.63	Beta-D-Glucopyranose,1,6-anhydro-	0.55	–	–
30	37.64	Cyclohexanecarbonitrile,1-hydroxy-	0.54	–	–
31	17.30	Catechol	0.51	2.24	–
32	14.96	1,2,4-Triazine-3,5(2H,4H)-dione	0.48	–	–
33	23.10	*meso*-5,5-Decanediol	0.43	–	–
34	16.51	1,2,4-Triazine-3,5(2H,4H)-dione	0.37	–	–
35	15.39	p-Heptyloxyaniline	0.35	–	–
36	12.52	2,5-pyrrolidinedione,1-methyl-	0.31	–	–
37	12.08	Hexanoic acid,2,7-dimethyloct-7-en-5-yn-4-yl ester	0.30	–	–
38	22.31	1-hexene,4-methyl-	0.29	–	–
39	19.10	2(3H)-Furanone,5-ethyldihydro-3-methyl-	0.27	–	–
40	13.84	p-Heptyloxyaniline	0.26	–	–
41	13.03	3-pyridinol	0.10	–	–
42	3.21	Benzene,1,3-bis(3-phenoxyphenoxy)-	–	10.36	58.93
43	11.80	p-Cresol	–	3.89	4.23
44	47.41	n-Hexadecanoic acid	–	3.73	2.99
45	31.38	2,4-Di-*tert*-butylphenol	–	3.57	
46	9.78	2-Cyclopenten-1-one, 2-Hydroxy-3-Methyl-	–	3.47	1.31
47	11.51	*cis*-4-Decenal	–	2.95	–
48	26.31	1-Tetradecene	–	2.11	–
49	13.34	Maltol	–	2.06	–
50	34.54	Cetene	–	1.97	0.67
51	16.63	5-(Hydroxymethyl)-Dihydro Furan-2-(3H)-one	–	1.72	–
52	10.94	Carbamic Acid, Methyl-, 3-Methylphenyl Ester	–	1.61	–
53	14.89	Glutarimide	–	1.50	–
54	52.77	9-Eicosenoic acid,(Z)-	–	1.43	–
55	20.88	Hydroquinone	–	1.30	–
56	8.77	Decane	–	1.27	–
57	10.10	Benzyl Alcohol	–	1.19	1.65
58	26.67	Tetradecane	–	1.14	–
59	4.86	Octanoic Acid	–	1.14	0.82
60	4.03	3-cyclopentene-1-acetaldehyde,2-oxo-	–	1.11	0.61
61	41.85	1-Heptadecene	–	1.06	–
62	34.83	Hexadecane	–	1.01	1.00
63	6.55	Pyridine, 3,4-Dimethyl	–	0.91	–
64	5.37	o-Xylene	–	0.88	–
65	38.17	1,4-Benzodioxin-6-propanoic acid,2,3-dihydro-	–	0.84	–
66	18.49	2,3-Anhydro-d-Mannosan	–	0.81	–
67	15.33	2-(1H)-Pyridinone, 3-Methyl-	–	0.77	–
68	24.06	Carbonic Acid, Dihexyl Ester	–	0.73	–
69	53.49	Octadecanoic acid	–	0.63	–
70	29.56	Pentanoic acid	–	0.46	–
71	26.98	3-Heptenoic acid	–	0.46	–
72	46.32	Proline	–	0.42	–
73	45.44	Hexadecanenitrile	–	0.37	–
74	33.37	Dodecanoic acid	–	0.36	–
75	54.47	Cyclohexane,1,1-(2-methyl-1,3-propanediyl)bis-	–	0.24	–
76	39.89	Pentadecane	–	–	2.02
77	6.08	1-Propene, 3-(Ethyleneoxy)-	–	–	1.28
78	20.60	Dodecane	–	–	1.20
79	61.41	Octanenitrile	–	–	1.17
80	20.10	2-Tetradecanol	–	–	0.87
81	27.15	Undecane	–	–	0.77
82	17.38	Benzene,1-methoxy-2-methyl	–	–	0.74
83	72.37	8-Heptadecene	–	–	0.73
84	9.03	2-Pentene, 1-Ethoxy-4,4-Dimethyl-	–	–	0.72
85	51.25	Decane,2,3–5-trimethyl	–	–	0.71
86	13.93	Phenol,2-methoxy	–	–	0.70
87	33.19	1-dodecanol	–	–	0.68
88	47.17	Benzenemethanol,α-phenyl-	–	–	0.60
89	6.93	Benzylamine	–	–	0.59

molecules would be subjected to fragmentation and lead to the rapid reduction in the levoglucosan content in the bio-oil (Leng et al., 2018). Additionally, there were many nitrogenous compounds present in the bio-oil, BO-300 and BO 400 such as amides, amines, however, their content was reduced significantly in BO-500 because the proteins in CFW underwent denitrogenating reaction and formed aromatic compounds in the bio-oil (Wang et al., 2021). Organic acids (carboxylic acids) were also present in bio-oils obtained at various pyrolysis temperatures. The acid fraction in bio-oil was increased by raising the temperature from 300 °C to 400 °C, because the degradation of cellulose, hemicellulose, and lignin would give rise to light oxygenates (carbonyl components that are maximum in BO-300), which would get oxidised into carboxylic acid (Sarchami et al., 2021). Some acidic compounds would become long chain aliphatic when the temperature increased to 500 °C, which would further fragment to produce gases or go through aromatization to create aromatic compounds.

The fraction of organic components in the BO-300, BO-400 & BO-500 that are present in>3% content (as per GC–MS analysis) were matched with the respective peaks of H-NMR,^{13}-C NMR& DEPT-NMR Analysis. Most of the peaks from the abovementioned analysis matched with the reference peaks of NMR analyses of respective components from the literature. The detailed component identification is mentioned in the supplementary material provided along with this paper.

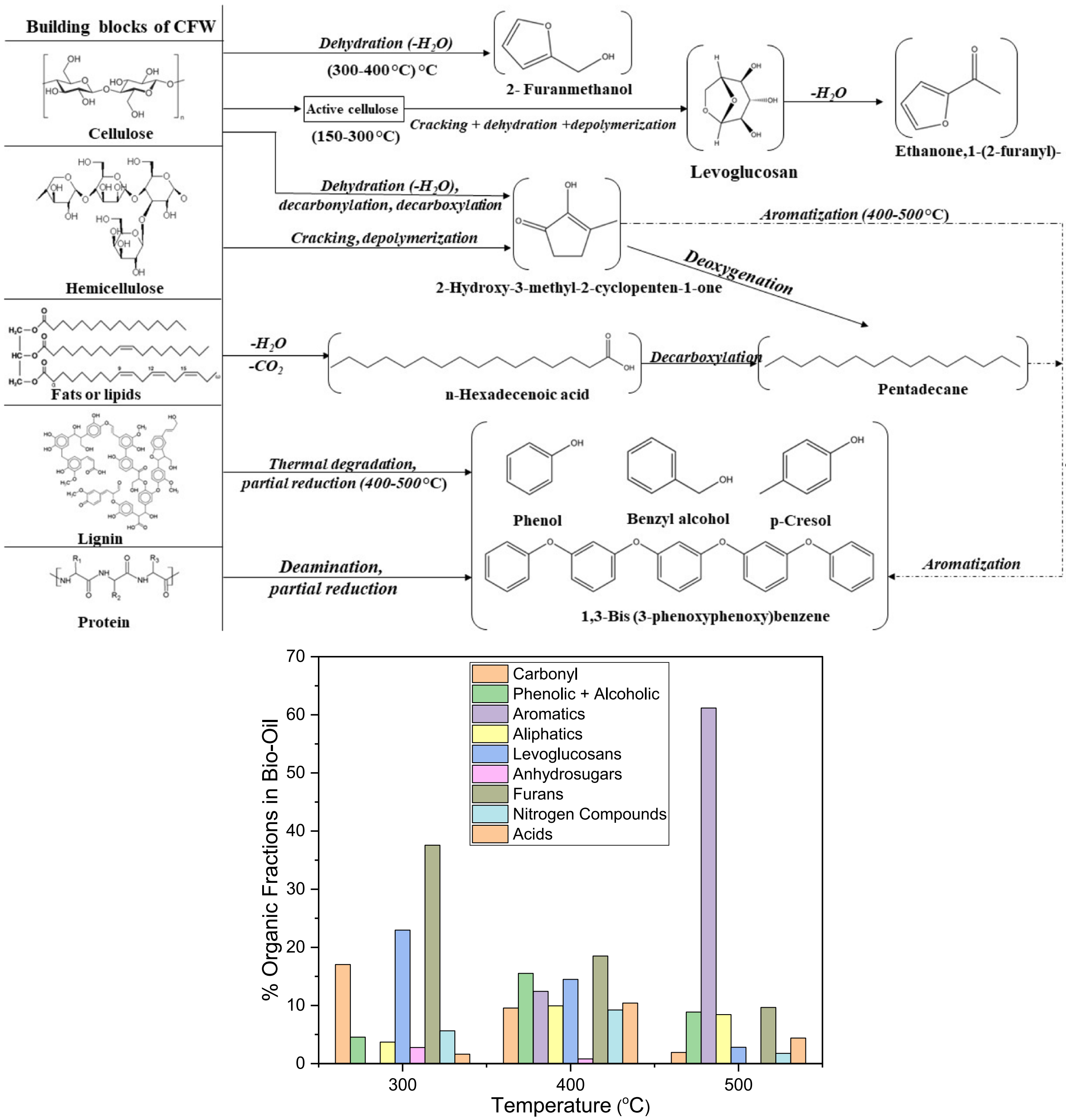


Fig. 4. (a): The proposed pyrolysis mechanism of CFW. Fig. 4(b): Fraction of various functional groups in bio-oil obtained at different temperatures.

3.8. Conclusions

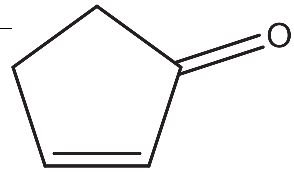
In this study, bio-oil was obtained from slow pyrolysis of CFW at three different temperatures: 300 °C, 400 °C, and 500 °C. The bio-oil yield increased significantly on increasing the temperature from 300 to 400 °C however a little decrease in bio-oil yield was noticeable with further increase in temperature up to 500 °C due to increase in syngas production. These bio-oil samples were characterized using NMR (H-NMR and ^{13}C NMR), and GC–MS analysis to check their molecular identity and composition. As per these analysis, Bio-oil of CFW contains multiple organic compounds such as aliphatics, aromatics, sugars, acids, phenolic and alcoholic compounds in the bio-oil and each component varied with changes in the pyrolysis temperature. H-NMR analysis showed the increase in methyl group protons and reduction in methoxy and carbohydrate protons in the bio-oil as the temperature increased from 300 °C to 500 °C.

Similarly, it showed the variation of protons of other compounds such as alkenes, alkanes, alcohols, and aromatics. ^{13}C NMR analysis of bio-oil samples showed the significant increase in long-chain aliphatic compounds and a drastic reduction in methoxy and carbohydrates with the increase in pyrolysis temperatures. Additionally, GC–MS analysis showed that there was a considerable increase in the aromatic compounds (~10% to 60 %) in the bio-oil obtained from pyrolysis at 500 °C. The furan compounds that were present in considerable amount in BO-300 and BO-400, were not present much (<10%) in BO-500. The compounds identified via GC–MS were confirmed via NMR plots of the compounds that were present in abundance of >3% in the bio-oil.

The high presence of aromatic compounds in BO-500 indicates toward the high quality of bio-oil generated by CFW pyrolysis at 500 °C as high presence of aromatics leads to higher stability of bio-oil. This

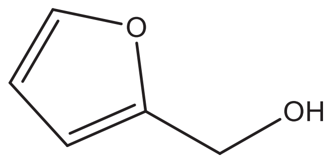
Table 4
DEPT confirmation of compounds identified by GC–MS.

(a) 2-Cyclopenten-1-one



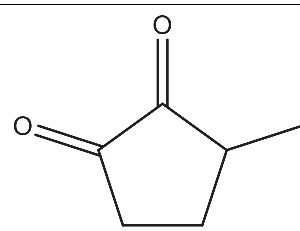
Peaks	Direction	Probable Groups	BO-300	BO-400	BO-500
29 ppm	Negative	CH_2	Yes	No	No
34 ppm	Negative	CH_2			
134 ppm	Positive	CH			
210 ppm	Positive	CH			

(b) 2-Furanmethanol



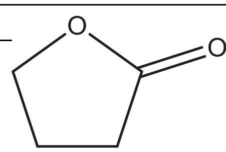
Peaks	Direction	Probable Groups	BO-300	BO-400	BO-500
57 ppm	Negative	CH_2	Yes	Yes	Yes
107 ppm	Positive	CH			
110 ppm	Positive	CH			
142 ppm	Positive	CH			
154 ppm	Positive	CH			

(c) 3-Methyl-1,2-cyclopentanedione



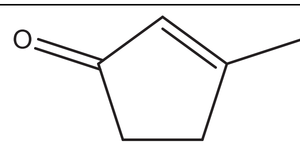
Peaks	Direction	Probable Groups	BO-300	BO-400	BO-500
15 ppm	Positive	CH	Yes	No	No
26 ppm	Positive	CH_3			
32 ppm	Negative	CH_2			
76 ppm	Negative	CH_2			
145 ppm	Positive	CH			
150 ppm	Positive	CH			
210 ppm	Positive	CH			

(d) Butyrolactone



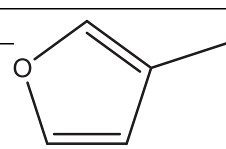
Peaks	Direction	Probable Groups	BO-300	BO-400	BO-500
23 ppm	Negative	CH_2	Yes	No	No
28 ppm	Negative	CH_2			
69 ppm	Negative	CH_2			
178 ppm	Positive	CH			

(e)3-Methyl-2-cyclopenten-1-one



Peaks	Direction	Probable Groups	BO-300	BO-400	BO-500
19 ppm	Positive	CH_3	Yes	No	No
33 ppm	Negative	CH_2			
36 ppm	Negative	CH_2			
137 ppm	Positive	CH			
178 ppm	Positive	CH			
208 ppm	Positive	CH			

(f) 3-Methyl Furan



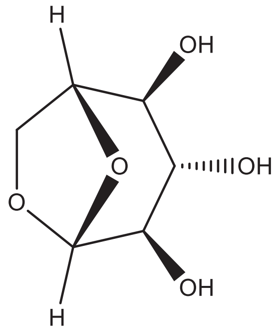
Peaks	Direction	Probable Groups	BO-300	BO-400	BO-500
9 ppm	Positive	CH_3	Yes	No	No
111 ppm	Positive	CH			
118 ppm	Positive	CH			
138 ppm	Positive	CH			
141 ppm	Positive	CH			

(g) Levoglucosan

Peaks	Directions	Probable Groups	BO-300	BO-400	BO-500
102 ppm	Positive	CH	Yes	Yes	No
72 ppm	Positive	CH			
73 ppm	Positive	CH			
71 ppm	Negative	CH_2			
76 ppm	Positive	CH			
64 ppm	Negative	CH_2			

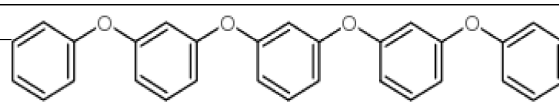
(continued on next page)

Table 4 (*continued*)



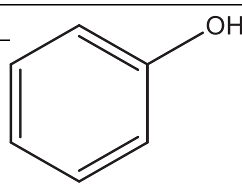
(h) Benzene,1,3-bis(3-phenoxyphenoxy)

Peaks	Directions	Probable Groups	BO-300	BO-400	BO-500
110 ppm	Positive	CH	No	Yes	Yes
120 ppm	Positive	CH			
130 ppm	Positive	CH			
152 ppm	Positive	CH			
158 ppm	Positive	CH			



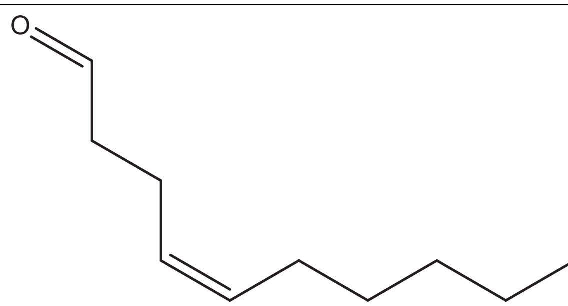
(i) Phenol

Peaks	Directions	Probable Groups	BO-300	BO-400	BO-500
117 ppm	Positive	CH	No	Yes	No
118 ppm	Positive	CH			
130 ppm	Positive	CH			
157 ppm	Positive	CH			



(j) Cis-4-Decenal

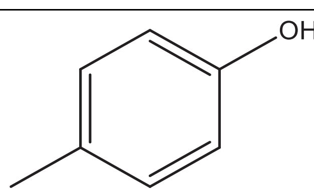
Peaks	Directions	Probable Groups	BO-300	BO-400	BO-500
14 ppm	Positive	CH	No	Yes	No
20 ppm	Positive	CH_3			
22 ppm	Negative	CH_2			
23 ppm	Negative	CH_2			
26 ppm	Negative	CH_2			
28 ppm	Negative	CH_2			
32 ppm	Negative	CH_2			
127 ppm	Positive	CH			
121 ppm	Positive	CH			



Cis-4-Decenal

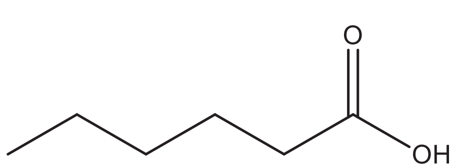
(k) p-Cresol

Peaks	Directions	Probable Groups	BO-300	BO-400	BO-500
22 ppm	Positive	CH	No	Yes	Yes
118 ppm	Positive	CH			
155 ppm	Positive	CH			



(l) n- Hexanoic Acid

Peaks	Directions	Probable Groups	BO-300	BO-400	BO-500
13 ppm‘	Positive	CH_3	No	Yes	No
22 ppm	Positive	CH			
24 ppm	Negative	CH_2			
31 ppm	Negative	CH_2			
34 ppm	Negative	CH_2			
180 ppm	Negative	CH_2			



implies that an improved quality of bio-oil can be produced from slow pyrolysis of CFW as compared to slow pyrolysis of typical biomass due to presence of additional compounds in CFW such as starch, proteins, waxes, oils and minerals. Furthermore, presence of aromatics in the form an ether (Benzene, 1,3-bis (3-phenoxyphenoxy)) in a very large fraction (58%) can be used to synthesize many chemicals using heterogeneous catalytic reactions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The authors would like to thank Monica Lohach (Research Scholar) for helping in carrying out GC-MS analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasman.2023.01.002.

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